Proj2

March 2, 2026

1 Project 2

2 Work From Home Burnout Analysis

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2.1 Data Overview

We used a WFH Burnout dataset. The data set tracks 150 employees over a total of 90 days. Includes about 2000 records in total over 13 weeks.

In this dataset, we are researching factors that may lead to a higher risk of burnout while working from home, and factors that may lower one's risk of burnout.

Goal: * What are the main factors that result in a higher risk of burnout? * And are there any suggestions based on the data to prevent burnout? * What factors lower someone's burnout risk? * Other factors that may raise one's burnout risk

2.2 Libraries

[1]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

In this analysis, Pandas will be used to create the dataframe and statistal analysis, while Seaborn and MatPlotLib will be used for graphing.

2.3 Loading the dataset

[2]:
```
df = pd.read_csv("wfh_burnout_dataset.csv")
```

Loading the WFH Burnout dataset.

2.4 Missing values check

[3]:
```
df.isna().sum()
```

[3]:
```
user_id              0
day_type             0
work_hours           0
screen_time_hours    0
meetings_count       0
breaks_taken         0
after_hours_work     0
app_switches         0
sleep_hours          0
task_completion      0
isolation_index      0
fatigue_score        0
burnout_score        0
burnout_risk         0
dtype: int64
```

2.5 Dataframe Preview and Summary

[4]:
```
df.head(6)
```

[4]:
```
   user_id day_type  work_hours  screen_time_hours  meetings_count  \
0      129  Weekday        7.74               5.99               3
1       24  Weekday        8.48               7.23               3
2      147  Weekend        8.39               7.14               1
3       42  Weekday       12.48              12.23               4
4       99  Weekend        8.85               7.35               4
5       54  Weekday        9.43               7.68               2

   breaks_taken  after_hours_work  app_switches  sleep_hours  task_completion  \
0             7                 0            50         7.50            74.80
1             5                 0            63         7.31            86.36
2             5                 0            27         6.73            89.23
3             1                 1            89         6.55            64.47
4             6                 0            40         8.03            98.62
5             7                 0            41         6.78            70.52

   isolation_index  fatigue_score  burnout_score burnout_risk
0                4           3.51          17.23          Low
1                4           5.15          24.01          Low
2                8           7.85          51.46       Medium
3                5          10.00          55.50       Medium
4                3           3.45          11.95          Low
5                5           5.83          34.58       Medium
```

[5]:
```
df.describe()
```

[5]:

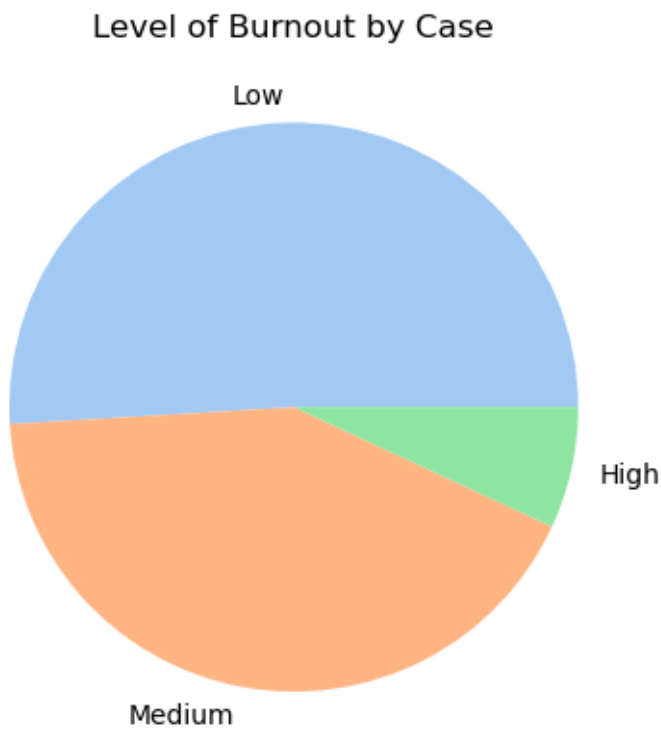
	user_id	work_hours	screen_time_hours	meetings_count	\
count	2000.000000	2000.000000	2000.000000	2000.000000	
mean	75.028000	8.955995	7.689495	3.728500	
std	43.317138	1.724214	1.983181	2.874877	
min	1.000000	4.730000	2.910000	0.000000	
25%	37.000000	7.750000	6.280000	2.000000	
50%	77.000000	8.535000	7.290000	3.000000	
75%	112.000000	9.980000	8.882500	5.000000	
max	150.000000	14.230000	13.980000	17.000000	

	breaks_taken	after_hours_work	app_switches	sleep_hours	\
count	2000.000000	2000.00000	2000.000000	2000.000000	
mean	5.066000	0.15650	56.025500	6.894900	
std	1.762441	0.36342	23.702463	0.983893	
min	0.000000	0.00000	5.000000	3.580000	
25%	4.000000	0.00000	39.000000	6.320000	
50%	5.000000	0.00000	52.000000	7.080000	
75%	6.000000	0.00000	71.000000	7.570000	
max	11.000000	1.00000	152.000000	9.220000	

	task_completion	isolation_index	fatigue_score	burnout_score
count	2000.000000	2000.000000	2000.000000	2000.000000
mean	80.090670	4.875500	6.560825	35.656590
std	12.708583	1.656011	2.218042	17.315039
min	40.000000	3.000000	1.450000	6.060000
25%	71.672500	3.000000	4.740000	21.005000
50%	80.900000	5.000000	6.295000	32.330000
75%	89.650000	6.000000	8.480000	47.475000
max	100.000000	9.000000	10.000000	82.220000

2.6 Graphical Overview of Burnout Risk

[6]:
```
colors = sns.color_palette('pastel')[0:3]
counts = df["burnout_risk"].value_counts()

plt.pie(counts.values, labels=counts.index, colors=colors)
plt.title("Level of Burnout by Case")
plt.show()
```



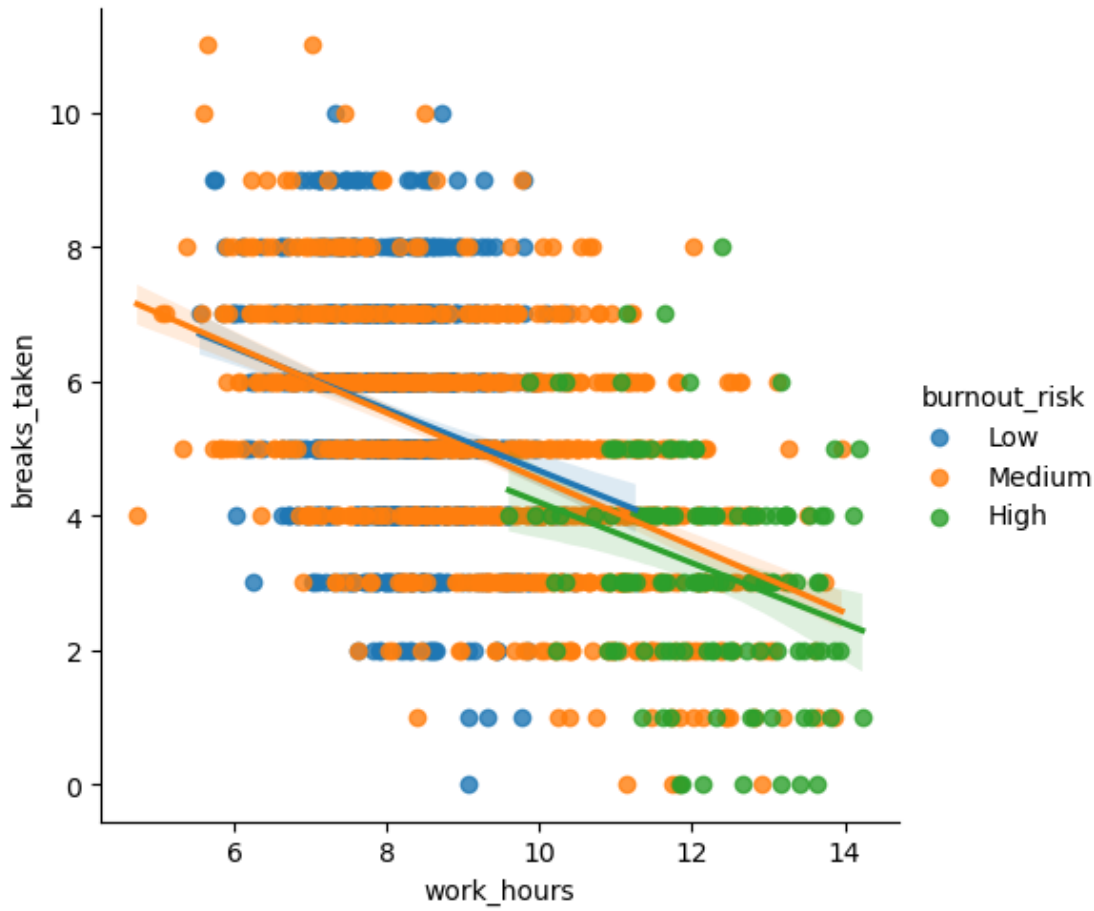
The pie chart above shows how skewed the data is, with the 'Low' category equal to the 'Medium' and 'High' categories. To counteract this and to ensure a fair analysis, a sample of each category will be drawn so that there is an even number in each category.

There's not much information that can be gathered from the plot above. The plot just gives an excellent numerical representation of how the data is split up.

2.7 Scatterplot - Clustering and Linear Regression

[7]:
```
sns.lmplot(data=df, x="work_hours", y="breaks_taken", hue="burnout_risk")
```

[7]:
```
<seaborn.axisgrid.FacetGrid at 0x23b2f996900>
```



In the above graph, we can see that our linear regression line runs almost evenly with the other categories and can just ablout determine that the number of hours and the number of breaks an employee takes has a direct effect on whether an employee has a low, medium, or high risk rate of burnout.

We can also see from the graph that everywhere there is a low risk or high risk, employees with a medium risk of burnout may overlap. So the question now becomes, what raises or lowers risk for those medium-risk employees?

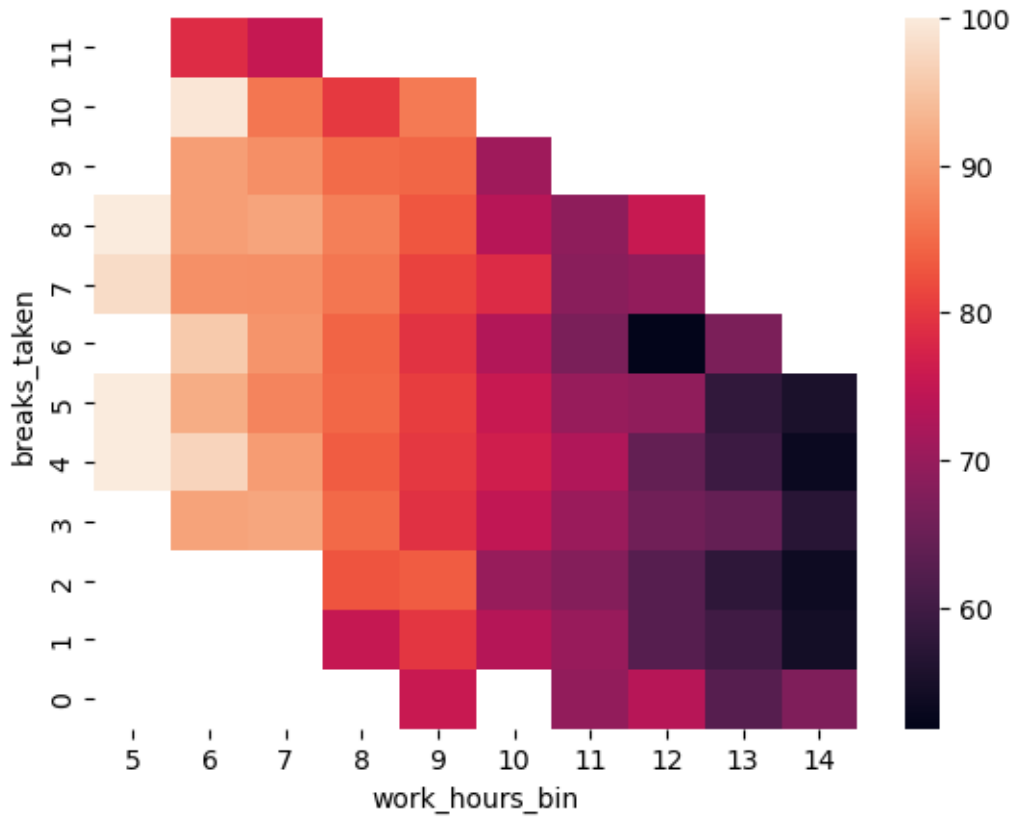
[8]:
```
slope, intercept = np.polyfit(df["work_hours"], df["breaks_taken"],  deg=1)
slope
```

[8]:
```
np.float64(-0.5152893668056232)
```

The above code snippet shows that the linear regression slope is estimated to be -0.52. This would mean that for every 1 less break that is taken and 2 more hours worked, both prodeuctivity and higher risk of burnout occurs within employees.

2.8 Productivity Heatmap

[9]:
```
df["work_hours_bin"] = df["work_hours"].round().astype(int)

prod = df.pivot_table(columns="work_hours_bin", index="breaks_taken",␣
↪values="task_completion", aggfunc="mean")


sns.heatmap(prod).invert_yaxis()
```



Due to MatPlotLib and Seaborn not supporting 3d graphs, I used a heatmap to best represent how as an employee does take more breaks and works less hours, there productivity and task completion also increases. As seen in the graph, the lighter the color, the more prductive that employee is for that week. And as we add more hours and less breaks for that employee there productivity also decreases.

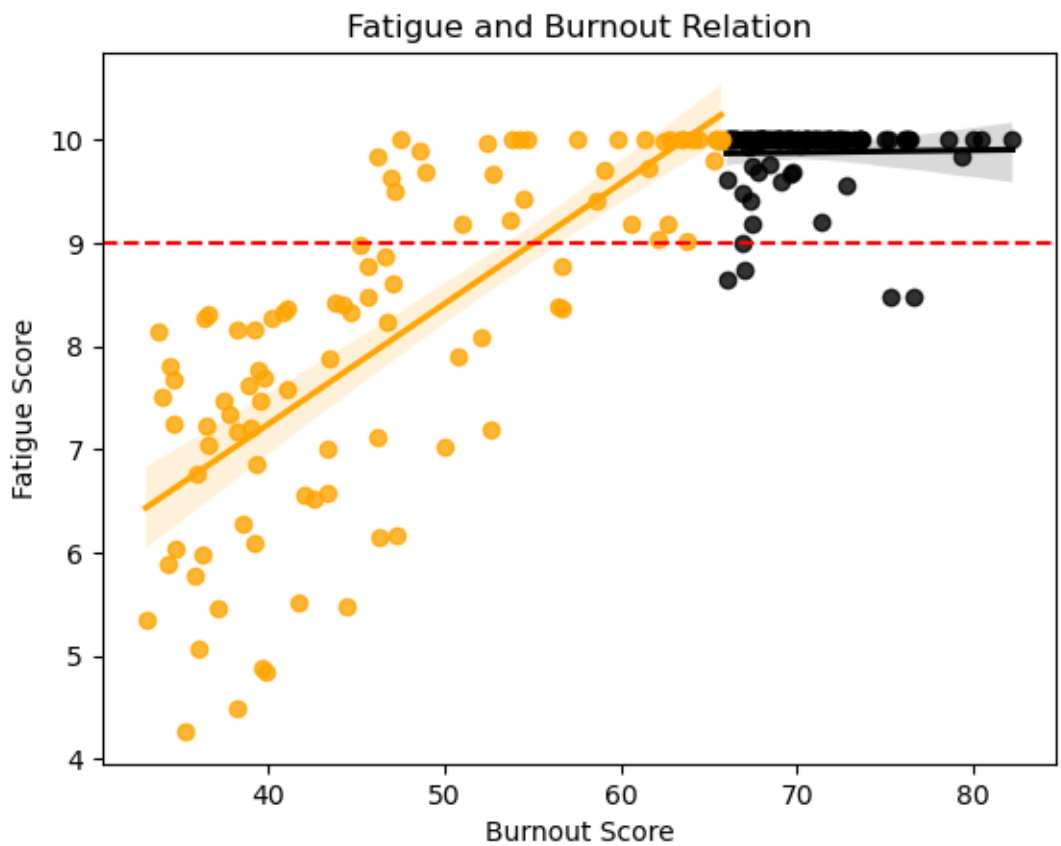
2.9 Filtering and Sampling

In the code below, samples are made for each burnout category to better level out the data population and take a deeper dive in the data.

[10]:
```
df_high = df[df["burnout_risk"] == "High"]
df_med = df[df["burnout_risk"] == "Medium"]
df_low = df[df["burnout_risk"] == "Low"]

samp_high = df_high.sample(n=100, random_state=4)
samp_med = df_med.sample(n=100, random_state=17)
samp_low = df_low.sample(n=100, random_state=26)
```

2.10 Fatigue Score Vs. Burnout Score

[11]:
```
sns.regplot(data=samp_high, x="burnout_score", y="fatigue_score", color="black")
sns.regplot(data=samp_med, x="burnout_score", y="fatigue_score", color="orange")
plt.axhline(y=9, color="red", linestyle='--')
plt.xlabel("Burnout Score")
plt.ylabel("Fatigue Score")
plt.title("Fatigue and Burnout Relation")
plt.show()
```



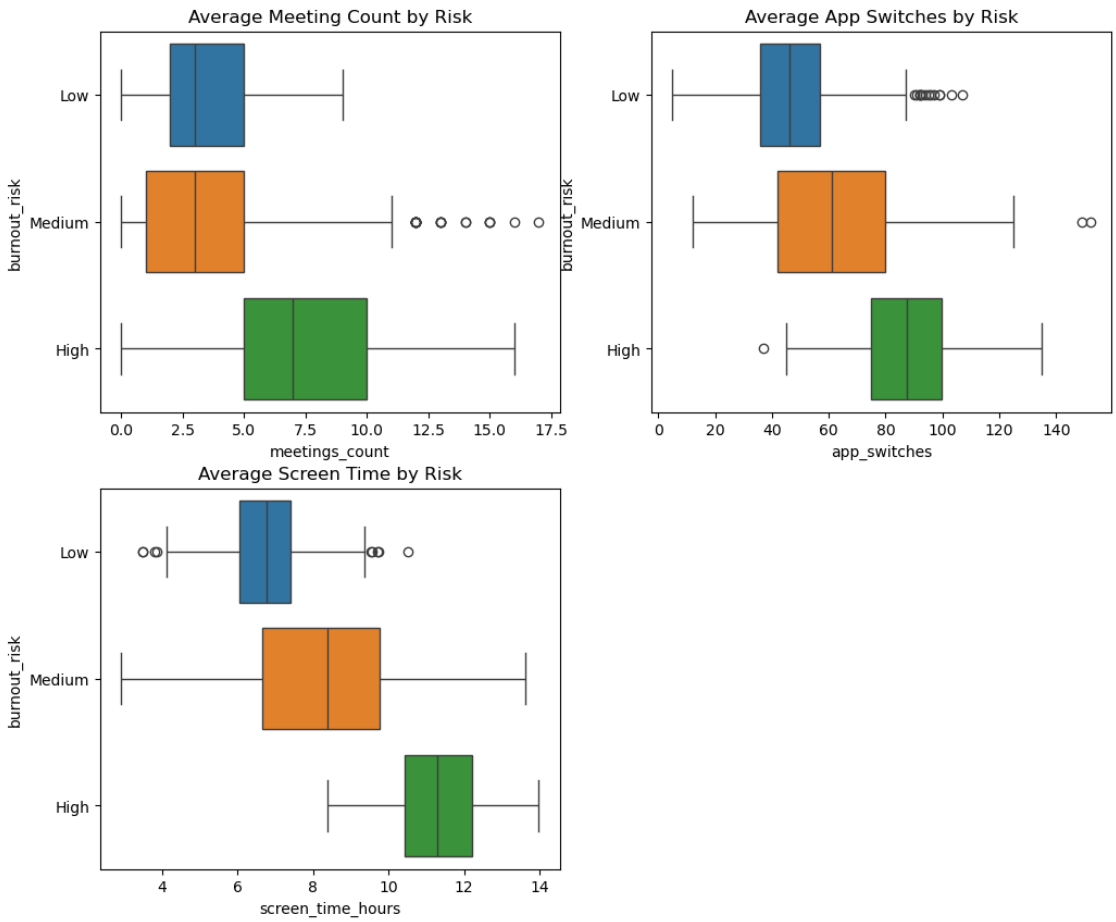
With orange representing medium burnout and black representing high burnout risk, we can see from the graph above. As the fatigue score rises, one can see that the change to a higher burnout

risk becomes more probable. We could set a line at around nine to make that the threshold for those with a higher chance of risking of burnout. We could also make a conclusion that those that are under the nine threshold are the reason that the medium burnout employees aren't in the high category.

2.11 Average Number of Meetings and Hourly Screen Time

[12]:
```
fig, axes = plt.subplots(2, 2, figsize=(12,10))

sns.boxplot(data=df, x="meetings_count", y="burnout_risk", hue="burnout_risk",␣
 ↪ax=axes[0,0])
axes[0,0].set_title("Average Meeting Count by Risk")

sns.boxplot(data=df, x="screen_time_hours", y="burnout_risk",␣
 ↪hue="burnout_risk", ax=axes[1,0])
axes[1,0].set_title("Average Screen Time by Risk")

sns.boxplot(data=df, x="app_switches", y="burnout_risk", hue="burnout_risk",␣
 ↪ax=axes[0,1])
axes[0,1].set_title("Average App Switches by Risk")

mean_1 = df.groupby("burnout_risk")["meetings_count"].mean().round(1)
mean_2 = df.groupby("burnout_risk")["screen_time_hours"].mean().round(1)
mean_3 = df.groupby("burnout_risk")["app_switches"].mean().round(1)

axes[1,1].axis("off")
```

[12]:
```
(np.float64(0.0), np.float64(1.0), np.float64(0.0), np.float64(1.0))
```



From the visual data above, we can see that most of the data can shows a progressive line. If we look at the 'App Switches' and the 'Screen Time' box plots. We could take a line and draw through the intersection and see that as the numbers increase the risk also increases. As for the 'Meeting Count' box plot, the average for both low and medium risk is equivalent and the immediate range of the box plot ends at the same point. Which can be said that the threshhold before an employee is at chance for high burnout risk is about five daily meetings.

What is guaranteed form the data is that an employee should limit their meetings to five per day.

[13]:
```
print(f"Average meeting count by risk:\n\n{mean_1}\n\n")
print(f"Average app switches by risk:\n\n{mean_2}\n\n")
print(f"Average screen time by risk:\n\n{mean_3}\n\n")
```

```
Average meeting count by risk:

burnout_risk
High      7.5
Low       3.4
Medium    3.5
Name: meetings_count, dtype: float64
```

```
Average app switches by risk:

burnout_risk
High      11.3
Low        6.8
Medium     8.2
Name: screen_time_hours, dtype: float64


Average screen time by risk:

burnout_risk
High      87.3
Low       46.6
Medium    62.3
Name: app_switches, dtype: float64
```

2.12 In Conclusion

In conclusion, as seen throughout the data, there are factors that can be controlled that cause burnout in the work from home workplace. In almost every plot, a trend line can bre created to see predict what will happen if an employee does action "X" or action "Y" this amount of times. High burnout risk is preventable, and as we have seen in the data, lower burnout provides higher work productivity. Based on the findings, employers should promote ways that an employee can improve their work-life balance.

[14]:
```
display(df)
```

```
      user_id day_type  work_hours  screen_time_hours  meetings_count  \
0         129  Weekday        7.74               5.99               3
1          24  Weekday        8.48               7.23               3
2         147  Weekend        8.39               7.14               1
3          42  Weekday       12.48              12.23               4
4          99  Weekend        8.85               7.35               4
…       …       …          …                  …              …
1995      101  Weekend        7.93               6.43               1
1996      114  Weekend        8.37               6.87               5
1997       49  Weekend       10.97              10.47              13
1998        9  Weekday        8.48               6.98               0
1999      149  Weekend        8.04               7.04               7

      breaks_taken  after_hours_work  app_switches  sleep_hours  \
0                7                 0            50         7.50
1                5                 0            63         7.31
```

```
2                5                  0             27          6.73
3                1                  1             89          6.55
4                6                  0             40          8.03
…              …                  …             …           …
1995             6                  0             22          7.75
1996             6                  0             89          7.00
1997             2                  0             93          5.86
1998             6                  0             29          8.12
1999             4                  0             32          7.62

      task_completion  isolation_index  fatigue_score  burnout_score  \
0               74.80                4           3.51          17.23
1               86.36                4           5.15          24.01
2               89.23                8           7.85          51.46
3               64.47                5          10.00          55.50
4               98.62                3           3.45          11.95
…               …                  …             …             …
1995            73.67                4           2.57          13.16
1996            80.05                5           7.13          35.41
1997            46.95                6          10.00          58.79
1998            72.57                6           6.41          32.64
1999            85.88                5           4.61          24.13

     burnout_risk  work_hours_bin
0             Low               8
1             Low               8
2          Medium               8
3          Medium              12
4             Low               9
…           …               …
1995          Low               8
1996       Medium               8
1997       Medium              11
1998          Low               8
1999          Low               8

[2000 rows x 15 columns]
```

2.13 Sources

https://www.kaggle.com/datasets/amar5693/screen-time-sleep-and-stress-analysis-dataset